THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A person who has run fast feels muscle pain due to: 1

(a) lactic acid accumulation **(b)** CO_2 shortage
(c) glucose excess **(d)** ethanol accumulation

2. Loss of water from leaves creates: 1

(a) blood pressure **(b)** osmosis only
(c) root pressure only **(d)** transpiration pull

3. Stomata are mostly closed at night because the guard cells: 1

(a) lose water and become flaccid **(b)** turn into root hairs
(c) take in water and swell **(d)** disappear

4. The breakdown of glucose to pyruvate occurs in: 1

(a) mitochondria **(b)** plastid **(c)** nucleus **(d)** cytoplasm

5. Water moves into root hair cells by: 1

(a) photosynthesis **(b)** translocation
(c) diffusion of gases **(d)** osmosis

6. Pancreatic juice contains: 1

(a) HCl **(b)** trypsin and lipase
(c) pepsin only **(d)** bile

7. During inhalation, the diaphragm: 1

(a) relaxes and bulges upward **(b)** becomes thick
(c) remains same **(d)** flattens

8. The rate of transpiration increases with: 1

(a) closing of stomata **(b)** decrease in temperature
(c) increase in humidity **(d)** increase in temperature

9. The inner lining of the stomach is protected from HCl by: 1

(a) amylase **(b)** mucus **(c)** bile **(d)** water

10. In fishes, gaseous exchange occurs through: 1

(a) gills **(b)** trachea **(c)** lungs **(d)** skin only

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between inhalation and exhalation. 2

12. Why are the arteries thick-walled? 2

13. Why is bile necessary for fat digestion though it has no enzyme? 2

14. Why is the rate of breathing in aquatic animals faster than in terrestrial ones? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the role of the villi in the small intestine. 3

16. Compare the structure of an artery, a vein and a capillary. 3

17. How is the carbon dioxide transported and removed from the body? 3

OR

Why does anaerobic respiration release less energy than aerobic respiration? 3

18. Explain how water moves from root to leaf in a tall tree. 3

OR

Explain why plant respiration is different from animal respiration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how oxygen is transported and exchanged in the lungs and the tissues. 5

OR

Explain the role of blood components: plasma, RBC, WBC, platelets, lymph. 5

20. Explain double circulation with a flow chart and its advantage. 5

OR

Explain how a plant gets rid of wastes. Compare with humans. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND